

S2CHANGE: Desenvolvimento de mapas de perdas recentes de floresta e mato em Portugal derivados de imagens de satélite

Contrato N.º 3044 de cooperação entre a Direção Geral do Território e o Instituto Superior de Agronomia

Entregável 4.1. Manual de utilização operacional da metodologia implementada na cadeia de produção da DGT.

(O relatório é redigido em Inglês)

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ISA_DGT_INCD_DL_pipeline is the computation pipeline developed at the INCD HPC cluster, operated by LIP, to process complete time series of Sentinel-2 imagery for Portugal. The goal of ISA_DGT_INCD_DL_pipeline is to produce near-real-time national maps of vegetation loss for areas of forest, bushes or shrubs.

Native data

The input data consists of:

1. (NativeS2) Sentinel-2 full time series for Portugal, 16-bit, 10 bands, 10 m resolution, 17 tiles of approximately 110 km by 110 km.

2. PT_mask: Vector file that represents the boundary of Continental Portugal, with a 2 km buffer
3. (DGT_mask) DGT vegetation loss mask in vector format for potential areas (forest, bushes or shrubs) where vegetation loss can occur. It covers approximately 50'000 km2 of the Portuguese territory
4. Reference data for training/validation/test – see “Entregável 4.3” and https://github.com/S2change/vegetation_loss/tree/main/data_info/reference_data

Pre-processed Sentinel-2 spectral data (HDF5 files)

The native Sentinel-2 Surface Reflectance imagery is acquired by CNCA in 16-bit geotiff format, with possibly more than one geotiff file per day since each combination of orbit and tile is recorded in one or more distinct geotiff files. Cloud-cover is estimated by Copernicus per tile and orbit, not taking into account the particular national cover.

The geotiff files were pre-processed with OMNI Cloud Mask (<https://omnicloudmask.readthedocs.io>) by LIP. This produced a cloud cover mask for the Portuguese territory (NoData=65535).

Then, those geotiff files with the OMNI mask are filtered, aggregated per day, and converted into hdf5 format to speed-up reading for further processing. They are filtered based on the estimated cloud cover over the Portugueses territory (PT_mask), with daily aggregated images being discarded is cloud cover over PT_mask is higher than 60% . Details can be found at https://github.com/S2change/vegetation_loss/blob/main/scripts/data_exploration/tifs_to_hdf5_to_tifs/CNCA_tifs_to_hdf5/readme.md

HDF5 files available at CNCA will be continuously updated at CNCA with new Sentinel-2 acquisitions following the same protocol.

Availability of data in hdf5 format allows fast access to temporal series of dates for any location. Here's a list of some of the main features of the hdf5 files available at the INCD HPC cluster:

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Root-level attributes

- band_names: ['B2' 'B3' 'B4' 'B5' 'B6' 'B7' 'B8' 'B8a' 'B11' 'B12']
- chip_size: 256
- crs: EPSG,"32629"
- nodata_val: 65535
- pixel_res: 10

HDF datasets:

- [S2_filename]
- [S2_original_filenames]

- [clear_pixel_count_pt]
- [cloud_cover_pt]
- [count_orbit_pixels_pt]
- [original_timestamps] milliseconds
- [pixel_count_pt]
- [ts] date, in ordinal format
- [values] ← spectral cube
 - - Shape : (191, 10, 102891520)
 - - Dtype : uint16
 - - Chunks : (12, 10, 65536)
 - Compression: lzf
 - Fill value : 65535
- [xs_new] x coordinates, EPSG: 32629"
- [ys_new] y coordinates

TEMPORAL COVERAGE

- Timestamp dataset: 'original_timestamps'
- Unit : milliseconds since Unix epoch
- N timestamps
- Date range

SPECTRAL & SPATIAL SUMMARY for Spectral dataset : 'values'

- Timestamps
- Bands : 10
- Pixels per date : varies, e.g. 102,891,520
- Data type : uint16 (uint16 range 0 – 65534; 65535 = NoData)
- NoData sentinel : 65535
- Chunk layout : 12 timestamp(s) × 10 band(s) × 65536 pixel(s)
- Per-timestamp metadata datasets: cloud_cover_pt, clear_pixel_count_pt, count_orbit_pixels_pt, S2_filename, S2_original_filenames

=====

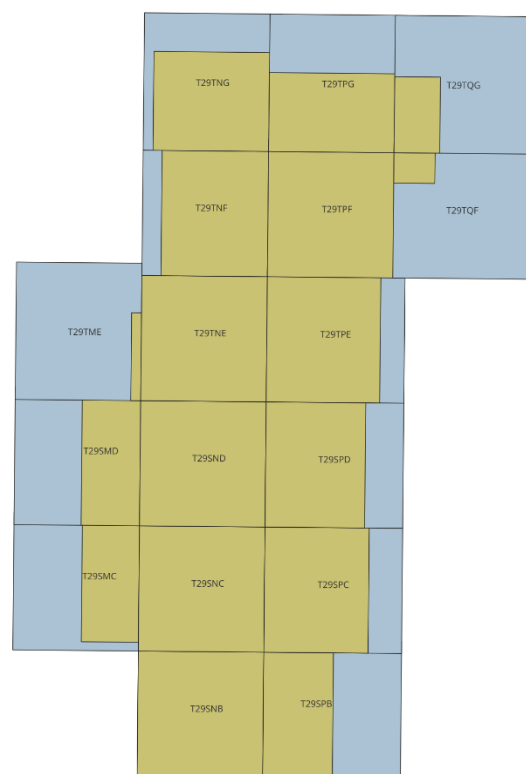


Figure 1: Sentinel-2 tiles for Portugal and minimal bounding boxes for each tile. There is a single multi-year hdf5 file for each tight bounding box (17 files in total).

The ISA_DGT_INCD_DL_pipeline

A key objective of the project was to develop a complete processing pipeline, spanning from input data to the generation of near-real-time National Vegetation Loss Maps, designed to run on the INCD HPC cluster.

Input hdf5 files are available at the INCD HPC cluster at

- users1/dgt/hdf5

The scripts for the pipeline are available at:

- INCD HPC cluster, at
/users1/cpca070342024/shared/vegetation_loss/scripts/pybacdm/cpca070342024_shared/predict_pipeline/,
- They are cloned from the following GitHub repository:
https://github.com/S2change/vegetation_loss/tree/main/scripts/cpca070342024_shared/predict_pipeline

The main batch file **submit_slurm.sh** executes the following steps for each tile:

1. input_setup to get blocks and dates
2. distributes a block to each array:

- a. runs `predict_block`
 - b. each array generates composites, shifted chips, predictions, votes, and returns results
3. `aggregate_tile.py` combines prediction results into full tile

Batch file **`submit_tiles_batch.sh`** submits the pipeline for many tiles in one command. It calls **`submit_tile.sh`** once per tile — each tile becomes its own independent SLURM array job + aggregator, so tiles run in parallel and SLURM spreads their block tasks across nodes. Per-tile load is still bounded by each submission's `MAX_CONCURRENT` cap, so submitting all tiles at once is safe (the scheduler queues whatever doesn't fit).

The script accepts a set of required and optional parameters, which are fully described in the accompanying

https://github.com/S2change/vegetation_loss/blob/main/scripts/cpca070342024_shared/predict_pipeline/README.md

Figure below illustrates how to submit a job for tile T29TNF over Sentinel-2 images with dates ranging from January 1, 2023 and February 15, 2024. The pipeline is launched on the HPC cluster by invoking the submission script `submit_tile.sh` as below:

```
[mlc@ui102 shared]$ submit_tile.sh TILE_ID=T29TNF
TILE_HDF5_PATH=/users1/dgt/hdf5/T29TNF.h5
OUTPUT_DIR=/users1/cpca070342024/shared/predict_outputs/00_test
START_DATE=2023-01-01 END_DATE=2024-02-15 MAX_COMPOSITE_DAYS=45
WRITE_COMPOSITE_TIFS=1 MAX_THETA=5 MAX_CLUSTER_AMPLITUDE=10
```

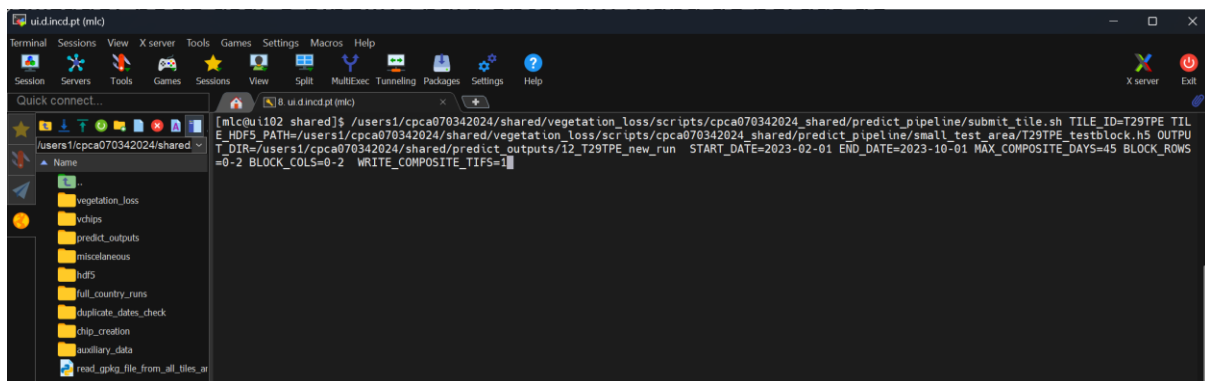


Figure 2: illustration of a run for tile T29TPE over Sentinel-2 images with dates ranging from February 1, 2023 to October 1, 2024

Currently, three alternative deep learning models are available to be called in the pipeline.

The choice can be set using option “MODEL” (which defaults to “efficientnet_b2_16bit_pipeline”). All models are grouped in

https://github.com/S2change/vegetation_loss/tree/main/scripts/cpca070342024_shared/predict_pipeline/models

Potential dates of change and date aggregation

ISA_DGT_INCD_DL_pipeline includes a deep learning classifier at the chip level, which consists of 256 by 256 pixels corresponding to an area on the ground of 2560 m by 2560 m. Each chip processing is performed for a given date D. For that date, a "before D/after D" pair of temporal composites of 10 band Sentinel-2 images with 10 m resolution are produced and used as inputs for the classification.

Since Sentinel-2 available dates have an irregular temporal distribution due to the irregularity of the orbits combined with cloud cover, the user needs to decide which potential dates of change (values of D in the description above) are to be analyzed.

The choice might be related to the time span of the analysis.

1. First scenario: the user might want to analyze a full year with, say, approximately 100 available Sentinel-2 daily images. To make the analysis more robust and faster, the user decides to group all dates that are separated by at most 10 days, in temporal clusters that span at most 15 days. This is illustrated in Figure 3.
2. Scenario 2: The user wants to analyze a shorter time span of, for instance, two summer months with many Sentinel-2 clear acquisition dates. To do a detailed temporal analysis, the user aggregates all dates that are separated by at most 3 days, in temporal clusters that span at most 5 days. This is illustrated in Figure 4.

The inputs that control those choices, e.g. `MAX_THETA=10` `MAX_CLUSTER_AMPLITUDE=15`, are called as arguments when invoking the submission script `submit_tile.sh`

The aggregation of Sentinel-2 bands is done with the “min” aggregating function for all bands, which mitigates haze, clouds, and NoData values.

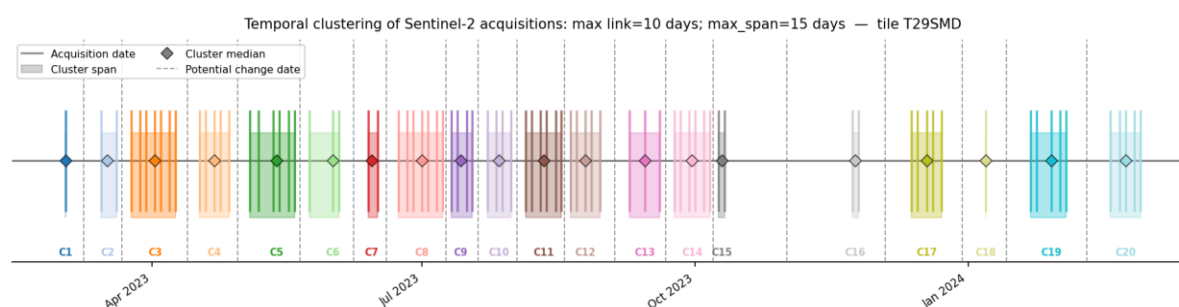


Figure 3: Temporal aggregation of dates separated by at most 10 days, in temporal clusters that span at most 15 days (parameters `MAX_THETA=10`; `MAX_CLUSTER_AMPLITUDE=15`). Colors represent different clusters of dates. Vertical colored lines represent daily Sentinel-2 observations. Dotted black vertical lines indicate potential vegetation loss dates to be analyzed.

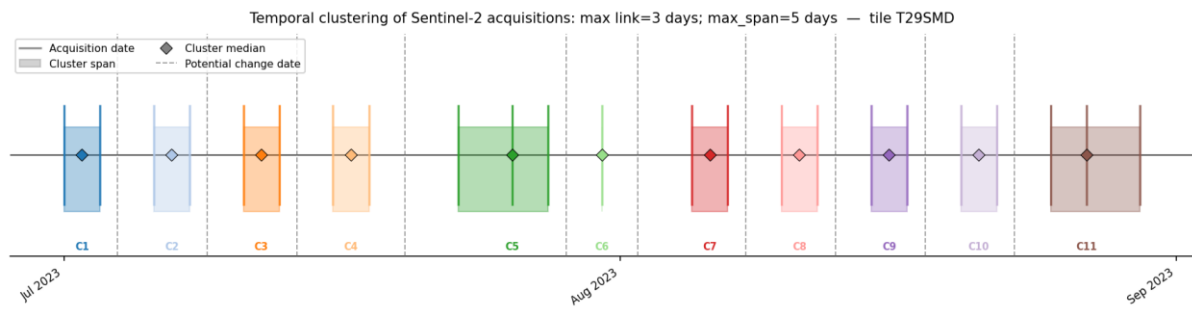


Figure 4: Temporal aggregation of dates separated by at most 3 days, in temporal clusters that span at most 5 days (parameters $MAX_THETA=3$; $MAX_CLUSTER_AMPLITUDE=5$). Colors represent different clusters of dates. Vertical colored lines represent daily Sentinel-2 observations. Dotted black vertical lines indicate potential vegetation loss dates to be analyzed.

Spatial extent

Since data is organized in 17 Sentinel-2 tiles (see Figure 1), tiles are a basic spatial unit when accessing data, which means that a spatial region that intersects more than 1 tile needs to be partition among tiles and processed in those parts.

However, the user is allowed to select only a few contiguous 10 km by 10 km blocks of a tile. When executing `submit_tile.sh` , the optional inputs that determine the blocks are `BLOCK_ROWS` / `BLOCK_COLS` (see description at https://github.com/S2change/vegetation_loss/blob/main/scripts/pybacdm/cpca070342024_shared/predict_pipeline/distribute/README.md).

Outputs

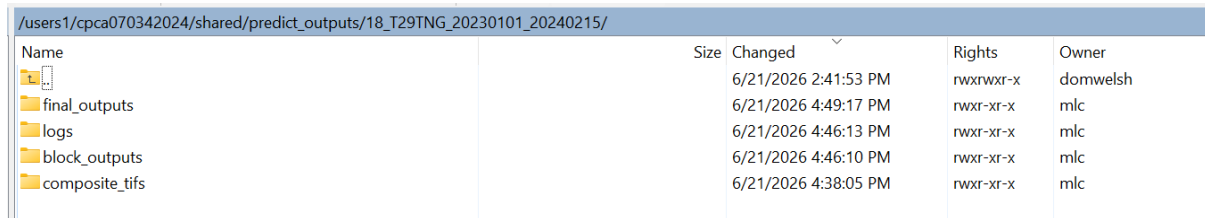
For the input tile and the optional spatial extent within the tile, and for the selected list of potential dates of change (see Section on Potential dates of change and date aggregation) the `ISA_DGT_INCD_DL_pipeline` will execute the following set of steps for each date `D`:

1. Construct from the aggregated date – illustrated in Figure 3 and Figure 4 -- a before `D` time composite (“before”), and an after `D` time composite (“after”) with 10 bands. Those are made of the valid (not NoData) spectral values at each pixel for date `D'` as close as possible from date `D`. The maximum range of dates $|D-D'|$ for “before” or “after” is set by parameter `MAX_COMPOSITE_DAYS`.
2. Create 256×256 spatial chips (with overlap) and apply the deep learning vegetation loss classification model to each chip. This outputs a label -- 0 for background (no change); 1 for “Cuts”; 2 for “Fires”) at the pixel level for each chip
3. Aggregate all chips and combine labels
4. Vectorize patches

For the set of potential dates of change, the pipeline combine all patches to create a geopackage file where features are the output patches and have attributes “`class_id`” (1 for

“Cuts”; 2 for “Fires”) and “data_iso” (in format yyyy-mm-dd) , which are the values D described above.

The outputs are written into folder which path is defined by parameter `OUTPUT_DIR`. If parameter `WRITE_COMPOSITE_TIFS=1`, then the output will include a subfolder with the time composites in geotiff format, as illustrated in Figure 5.



Name	Size	Changed	Rights	Owner
final_outputs		6/21/2026 2:41:53 PM	rw-rw-r--	domwelsh
logs		6/21/2026 4:49:17 PM	rw-r--r--	mlc
block_outputs		6/21/2026 4:46:13 PM	rw-r--r--	mlc
composite_tifs		6/21/2026 4:46:10 PM	rw-r--r--	mlc

Figure 5: Example of an output folder. The final geopackage file is available in folder “final_outputs”. Geotiff time composites are available in folder composite_tifs.

How to use

In short, the steps the user need to perform are the following:

1. Choose Sentinel-2 tile and, optionally, spatial 10 by 10 km blocks within the tile to analyze
2. Choose range of dates from the Sentinel-2 time series to read from hdf5 file. E.g.
`START_DATE=2023-01-01 END_DATE=2024-02-15`
3. Choose data aggregation parameters, e.g. `MAX_THETA=5`
`MAX_CLUSTER_AMPLITUDE=10`
4. Choose maximum range for time compositing , e.g. `MAX_COMPOSITE_DAYS=45`
5. Choose output folder path, e.g.
`OUTPUT_DIR=/users1/cpca070342024/shared/predict_outputs/00_test`
6. Consider other optional inputs described in
https://github.com/S2change/vegetation_loss/tree/main/scripts/cpca070342024_shared/predict_pipeline
7. Run command in terminal, e.g.

```
# All *.h5 in a directory:
/users1/cpca070342024/shared/vegetation_loss/scripts/cpca070342024_shared/predict_pipeline/submit_tiles_batch.sh
BASE_OUTPUT_DIR=/users1/cpca070342024/shared/predict_outputs/13_summer2023
TILE_DIR=/users1/dgt/hdf5/ START_DATE=2023-07-01 END_DATE=2023-09-15
USE_DATE_CLUSTERS=1 MAX_THETA=5 MAX_CLUSTER_AMPLITUDE=5
```

```
# An explicit subset (TILE_DIR defaults to /users1/dgt/hdf5/):
/users1/cpca070342024/shared/vegetation_loss/scripts/cpca070342024_shared/predict_pipeline/submit_tiles_batch.sh
BASE_OUTPUT_DIR=/users1/cpca070342024/shared/predict_outputs/13_summer2023
TILES="T29SNB T29TPE T29TPG" START_DATE=2023-07-01 END_DATE=2023-09-15
```

8. Get geopackage T29xxx.gpkg output from `OUT_DIR/final_outputs` when processing is complete

Computational resources

Below are listed some indicators of CPU time and wall-to-wall time:

1. Two month period, with parameters `MAX_THETA=5` `MAX_CLUSTER_AMPLITUDE=10`, corresponding to approximately 8 potential dates would need approximately 45 minutes of processing for the whole Portugal. Tiles are processed in parallel, so the total time depend mostly on the largest tile. For instance, for tile T29TNE
 - a. Total tile wall time (first block start -> aggregator finish): 00:46:52 (2812 s), tile T29TNE
 - b. Total tile CPU time (all blocks + aggregator): 07:44:46 (27886 s), tile T29TNE
2. Two year processing for a single tile (T29TNE) for 55 potential dates took approximately 4h20:
 - a. Total tile wall time (first block start -> aggregator finish): 04:17:00 (15420 s), tile T29TNE